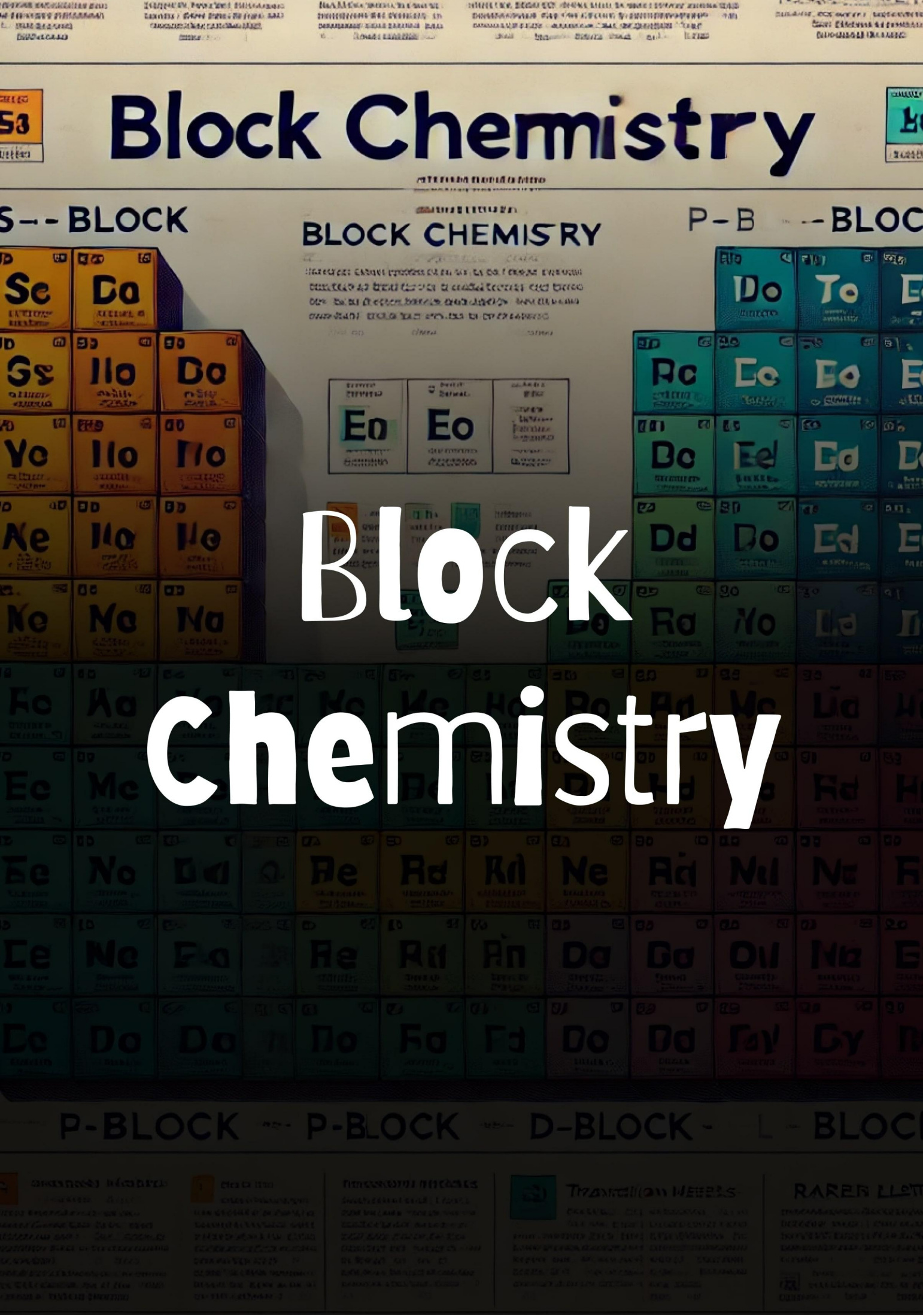

Block Chemistry
BLOCK CHEMISTRY
Block
Chemistry

GROUP - 1A

Electron negativity and affinity ↓ down the group.

Density - Li < Na < K < Rb < Cs (Vol. increase in "K")

Stability of superoxides down the group increases.

Cohesive energy density $\propto \frac{1}{size} \propto$ Hardness $\propto$ (mp, BP)

∴ Li > Na > K > Rb > Cs (softest)

REDUCING NATURE

In Non aqueous solvent :- Cs > Rb > K > Na > Li

In an aqueous solvent :- Li > Na > K > Rb > Cs

Hydration energy $\propto \frac{1}{size}$

All alkari metals dissolve in each other (except Li)

"Li" only dissolve in Magnesium

All alkari metals forms alum; except 'Li'

Potash alum → $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

chrome alum → $K_2SO_4 \cdot Cr_2(SO_4)_3 \cdot 24H_2O$

All alkari metals can be stored in kerosene; except 'Li' Lithium is a lights metal, stored in wax.

All alkali metals impart colour to the flame.

Li → Red

Na → golden Yellow

K → Violet

Rb → Red Violet

Cs → Blue

(Be, Mg) don't impart colour to the flame due to their small size and more energy is required to excite the e^- from the lower energy state to the higher energy state.

REACTION WITH O_2

$4Li + O_2 \longrightarrow 2Li_2O$

$Na + O_2 \longrightarrow Na_2O + Na_2O_2$

These gives superoxides:

$K + O_2 \longrightarrow K_2O + K_2O_2 + KO_2$

$Rb + O_2 \longrightarrow$ ✓

$Cs + O_2 \longrightarrow$ ✓

Stability of superoxides - $CsO_2 > RbO_2 > KO_2$

Peroxide is an oxidising agent.

$$2Na_2O_2 + 2H_2O \longrightarrow 4NaOH + O_2$$

(Na_2O_2) is used as a purifier and converts CO_2 in Na_2CO_3 but (KO_2) is a better purifier as it gives out more O_2.

Only Lithium reacts with "N_2" to form covalent nitride

$$6Li + N_2 \longrightarrow 2Li_3N \text{ (Lithium nitride)}$$

$$Li_3N + 3H_2O \longrightarrow NH_3 + 3LiOH$$

NH_3 + NaOCl → NH_2-NH_2

Only Lithium reacts with carbon to form covalent carbide

$$2Li + 2C \longrightarrow Li_2C_2 \text{ (covalent)}$$

$$Li_2C_2 + 2H_2O \longrightarrow 2LiOH + C_2H_2$$

Alkali metals react with H_2 to form hybrides (ionic)

$$2Li + H_2 \longrightarrow 2LiH \qquad 2Cs + H_2 \longrightarrow 2CsH$$

All alkali metals and hydrides of them give 'H_2' with H_2O (exothermic)

$$2M + H_2O \longrightarrow 2MOH + H_2\uparrow$$

REACTION WITH NH_3

$$M + NH_3(dil) \longrightarrow M(NH_2)_x + e^-_{(solvated\ e^-)}$$

($M(NH_2)_x$: Solvated metal ion)

Deep blue colour → Paramagnetic, conductive soln, reducing agent.

After sometimes the colour fades away due to the formation of amides and colour of the solution becomes bronze (Diamagnetic).

$$2Na + 2NH_3(dil) \longrightarrow 2NaNH_2 + H_2\uparrow$$

NaOH (Caustic soda)

By Castner's Kellner cell (use of brine solution).

H_2 is evolved at the cathode while Cl_2 at anode.

- $NaOH + Zn \longrightarrow ?$

$$Zn + 2H_2O \longrightarrow H_2ZnO_2 + H_2$$

$$H_2ZnO_2 + 2NaOH \longrightarrow Na_2ZnO_2 + 2H_2O$$

$$\boxed{Zn + 2NaOH \longrightarrow Na_2ZnO_2 + H_2}$$

Amphoteric metals will always give H_2 gas.

$PbO + NaOH \longrightarrow ?$

$PbO \longrightarrow Pb + (O)$

$Pb + 2H_2O \longrightarrow H_2PbO_2 + H_2$

$H_2 + (O) \longrightarrow H_2O$

$H_2PbO_2 + 2NaOH \longrightarrow Na_2PbO_2 + 2H_2O$

$$PbO + 2NaOH \longrightarrow Na_2PbO_2 + H_2O$$

Amphoteric metal oxides give H_2O

$Al(OH)_3 + NaOH \longrightarrow ?$

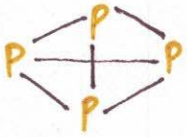
$Al(OH)_3 \longrightarrow H_3AlO_3$

$H_3AlO_3 + 3NaOH \longrightarrow Na_3AlO_3 + 3H_2O$

$$Al(OH)_3 + 3NaOH \longrightarrow Na_3AlO_3 + 3H_2O$$

WITH NON-METALS

only white P_4 reacts and it shows disproportionation



$+ 6H_2O \longrightarrow PH_3 + 3H_3PO_2$

(Red^n)

$3H_3PO_2 + 3NaOH \longrightarrow 3NaH_2PO_2 + 3H_2O$

(replaced H) ($oxid^n$) (sodium hypo-phosphite)

$$P_4 + 3NaOH + 3H_2O \longrightarrow PH_3 + 3NaH_2PO_2$$

(white)

$4S + 6NaOH \longrightarrow Na_2S + Na_2S_2O_3$

(first into lowest O.S then the next no.)

REACTION WITH X_2 – where $(Cl, Br, I) = X$

Cl_2 + cold and dil. NaOH

$Cl_2 + H_2O \longrightarrow HCl + HOCl$

$HCl + NaOH \longrightarrow NaCl + H_2O$

$HOCl + NaOH \longrightarrow NaOCl + H_2O$

$$Cl_2 + 2NaOH \longrightarrow NaCl + NaOCl + H_2O$$

(b) Cl_2 + hot and conc. NaOH

$$3Cl_2 + 3H_2O \longrightarrow HClO_3 + 5HCl$$

$$5HCl + 5NaOH \longrightarrow 5NaCl + 5H_2O$$

$$HCl_3 + NaOH \longrightarrow NaClO_3 + H_2O$$

$$3Cl_2 + 6NaOH \longrightarrow 5NaCl + NaClO_3 + 3H_2O$$

In fluorine, no disproportionation is possible as only one O.S is possible (O.S. = 1)

Reaction with salts that give soluble hydroxides in NaOH.

$$ZnCl_2 + NaOH \longrightarrow ?$$

similar for $Pb(NO_3)_2$, $AlCl_3$, $BeCl_2$

$$ZnCl_2 + H_2O \longrightarrow 2HCl + ZnO$$

$$ZnO + H_2O \longrightarrow H_2ZnO_2$$

$$2HCl + 2NaOH \longrightarrow 2NaCl + 2H_2O$$

$$H_2ZnO_2 + 2NaOH \longrightarrow Na_2ZnO_2 + 2H_2O$$

$$ZnCl_2 + 4NaOH \longrightarrow 2NaCl + Na_2ZnO_2 + 2H_2O$$

Reaction which give soluable hydroxides in H_2O

$$CaCl_2 + 2NaOH \longrightarrow Ca(OH)_2 + 2NaCl$$

same for $BaCl_2$, $SrCl_2$, $Sr(NO_3)_2$

Reaction with salts that give insoluble $\overline{OH}$

d-block elements give unsoluble $\overline{OH}$

$$MnSO_4 + 2NaOH \longrightarrow Na_2SO_4 + Mn(OH)_2 \downarrow \text{ (Light pink)}$$

$$FeCl_2 + 2NaOH \longrightarrow 2NaCl + Fe(OH)_2 \downarrow \text{ (Green)}$$

$$FeCl_3 + 3NaOH \longrightarrow 3NaCl + Fe(OH)_3 \downarrow \text{ (Rust green)}$$

$Ni(OH)_2 \downarrow$ (Green); $Ca(OH)_2 \downarrow$ (Pink); $Cr(OH)_3 \downarrow$ Dirty green

$$BiCl_3 + 3NaOH \longrightarrow 3NaCl + Bi(OH)_3 \downarrow \text{ (white)}$$

Reaction with metals that give metal oxides

(a) $2AgNO_3 + 2NaOH \longrightarrow 2NaNO_3 + \underset{\text{(Brown)}}{Ag_2O} + H_2O$

(b) $Hg_2(NO_3)_2 + 2NaOH \longrightarrow \underset{\text{(Black)}}{Hg_2O} + NaNO_3 + H_2O$

(c) $HgCl_2 + 2NaOH \longrightarrow 2NaCl + \underset{\text{(Yellow)}}{HgO} + H_2O$

$CrO_2Cl_2 + NaOH \longrightarrow ?$

$CrO_2Cl_2 + H_2O \longrightarrow 2HCl + CrO_3$

$CrO_3 + H_2O \longrightarrow H_2CrO_4$

$H_2CrO_4 + 2NaOH \longrightarrow Na_2CrO_4 + 2H_2O$

$2HCl + 2NaOH \longrightarrow 2NaCl + 2H_2O$

$\underset{+b}{CrO_2Cl_2} + 4NaOH \longrightarrow \underset{+b}{Na_2CrO_4} + 2NaCl + 2H_2O$

$SO_2Cl_2 + NaOH \longrightarrow ?$

$SO_2Cl_2 + H_2O \longrightarrow 2HCl + SO_3$

$SO_3 + H_2O \longrightarrow H_2SO_4$

$2HCl + 2NaOH \longrightarrow 2NaCl + 2H_2O$

$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$

$\underset{+b}{SO_2Cl_2} + 4NaOH \longrightarrow Na_2SO_4 + \underset{+b}{2NaCl} + 2H_2O$

Same for many salts like $SOCl_2$, $CrCl_3$, etc.

Na_2CO_3 (WASHING SODA) – $Na_2CO_3 \cdot 10H_2O$

Many factured in the solvay process. The raw materials are $CaCO_3$, NH_3 NaCl (brine).

In the middle of the process – NH_4Cl, $NaHCO_3$ is also produced.

we can also get ($CaCl_2 \cdot 6H_2O$) → Dehydrating agent

Products of solvay's process – $NHCO_3$, Na_2CO_3.

By-prod. of solvay's process – NH_3, CO_2 (They are used again)

$Ca(NO_3)_2 + Na_2CO_3 \longrightarrow CaCO_3\downarrow + 2NaNO_3$

Reaction with salts that give insoluble $\overline{OH}$

We get a basic metal carbonate

$Zn(NO_3)_2 + Na_2CO_3 \longrightarrow 2NaNO_3 + ZnCO_3 \cdot Zn(OH)_2$

similar reaction for d-block elements

E.g. $CuSO_4$, $MnSO_4$, $Pb(NO_3)_2$. $Zn(NO_3)_2$ etc.

$MgCl_2 + Na_2CO_3 \longrightarrow 2NaCl + MgCO_3 \cdot Mg(OH)_2$

(Basic Magnesium carbonate)

$AlCl_3 + Na_2CO_3 \longrightarrow Al_2(CO_3)_3 + 6NaCl$

$Al_2(CO_3)_3 \longrightarrow Al_2O_3 + 3CO_2$

$AlCl_3 + Na_2CO_3 \longrightarrow Al_2O_3 + 6NaCl + 6CO_2$

$NaHCO_3$ (Baking soda)

It exists in the polymer form → acidic

$MgCl_2 + 2NaHCO_3 \longrightarrow 2NaCl + Mg(HCO_3)_2 \xrightarrow{\Delta} MgCO_3 + H_2O + CO_2\uparrow$

$MgCO_3$ is a pure carbonate. we can thus differentiate between $NaHCO_3$ and Na_2CO_3 by using $MgCl_2$.

Hypo ($Na_2S_2O_3 \cdot 5H_2O$)

$Na_2SO_3 + S \longrightarrow Na_2S_2O_3$; $2Na_2S + 3SO_2 \longrightarrow 2NO_2S_2O_3 + S\downarrow$

$Na_2SO_3 + 2Na_2S + 3SO_3 \longrightarrow Na_2S_2O_3$

$2Na_2S + 4SO_2 \quad Na_2CO_3 \longrightarrow 3Na_2S_2O_3 + CO_2\uparrow$

$3Na_2S + 4SO_2 \quad 2NaOH \longrightarrow 3Na_2S_2O_3 + H_2O$

By Boiling 'S' in basic medium (NaOH) → $Na_2S_2O_3 + Na_2S$

Spring's reaction

$Na_2S + I_2 \longrightarrow 2NaI + S$

$Na_2SO_3 + S \longrightarrow Na_2S_2O_3$

$Na_2S + Na_2SO_3\ I_2 \longrightarrow 2NaI + Na_2S_2O_3$

$Na_2S_2O_3 \cdot 5H_2O \xrightarrow{\Delta} Na_2SO_4 + Na_2S_5$

$Na_2S_2O_3 + 2HCL \longrightarrow 2NaCl + H_2O + SO_2$

Mechanism $Na_2S_2O_3 + 2HCL \longrightarrow H_2S_2O_3 + 2NaCl$

$H_2S_2O_3 \xrightarrow{\Delta} H_2O + SO_2$

$Na_2S_2O_3 + 2HCL \longrightarrow 2NaCl + H_2O + SO_2\uparrow$

$Na_2S_2O_3$ with $AgNO_3$

(a) Excess $AgNO_3 \longrightarrow NaNO_3 + Ag_2S$ (Black)

(b) Excess $Na_2S_2O_3 \longrightarrow NaNO_3 + Na_3[Ag(S_2O_3)_2]$

$Na_2S_2O_3$ as a reducing agent

$Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$

(sodium tetra thionate)

$FeCl_3 + 3Na_2S_2O_3 \longrightarrow 6NaCl + Fe_2(S_2O_3)_3$

(Not R.A) (Pink)

$2FeCl_3 + Na_2S_2O_3 \longrightarrow 4NaCl + 2FeCl_2 + Na_2S_4O_6$

(R.A) (Not Pink)

$CuCl_2 + 2Na_2S_2O_3 \longrightarrow 4NaCl + Cu_2Cl_2 + Na_2S_4O_6$

$CuCl_2 + 5Na_2S_2O_3 \longrightarrow Na_4[Cu_6(S_2O_3)_5]$
(excess) — 4,5,6 complex

Cl_2, Br_2 are strong O.A. They oxidise $Na_2S_2O_3$ into Na_2SO_4

$Cl_2 + H_2O + Na_2S_2O_3 \longrightarrow Na_2SO_4 + HCl + (S)$

$Br_2 + H_2O + Na_2S_2O_3 \longrightarrow Na_2SO_4 + HBr + (S)$

$AgBr$ (Yellow) $+ 2Na_2S_2O_3 \longrightarrow Na_3[Ag(S_2O_3)_2]$
soluble complex

$AgCl$ (Dark Yellow) $+ 2Na_2S_2O_3 \longrightarrow Na_3[Ag(S_2O_3)_2]$

$Na_2S_2O_3$ (anti-chlor) is used for removing chlorine. It is used in photography; AgBr dissolver in hypo.

$NaNO_2$ (Sodium nitate)

$N_2O_3 + NaOH \longrightarrow 2NaNO_2 + H_2O$

$NaNO_3 + Zn \longrightarrow NaNO_2 + ZnO$

$NaNO_3 + C \longrightarrow NaNO_2 + CO$

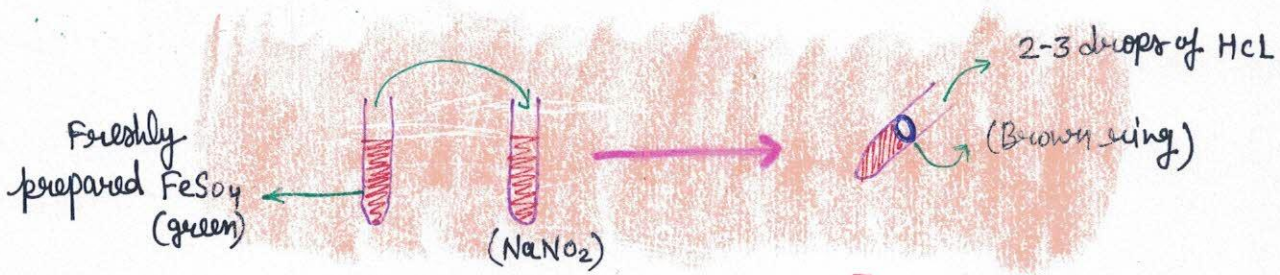
$4NaNO_2 \xrightarrow{(>800^\circ C)} Na_2O + 2N_2 + 3O_2$

$NaNO_2 + HCl \longrightarrow NaCl + 2NO\uparrow + HNO_3$

$NaNO_2 + H_2SO_4 \longrightarrow Na_2SO_4 + NO\uparrow + HNO_3$
(excess)

$NaNO_2 + H_2SO_4 \longrightarrow NaHSO_4 + 2NO\uparrow + HNO_3$
(excess)

BROWN RING TEST



$[Fe(H_2O)_5NO]^+ SO_4^{-2}$ → Brown ring

$[FeSO_4 \cdot NO]$ → (Nitroso ferrous sulphate)

$2NaNO_2 + 2H_2SO_4 + FeSO_4 \longrightarrow Na_2SO_4 + Fe_2(SO_4)_3 + 2FeSO_4 \cdot NO + H_2O$

Oxides of Potessium

K_2O ⟶ Diamagnetic (colourless)
K_2O_3 ⟶ Diamagnetic (colourless) ⟶ $K^+ \; O=O^- - O^- \; K^+$
K_2O_2 ⟶ Diamagnetic (colourless)
K_2O_2 ⟶ Paramagnetic

Na^+ - Transmission of nerve signals
K^+ ions are most abundant cations within the cell fluid ; where they activate enzyme Participate in the oxidation of glucose to produce ATP.

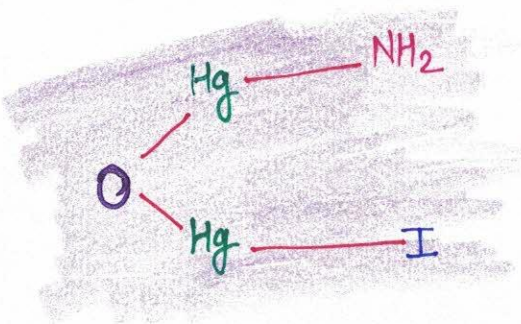
$$KOH + O_3 \longrightarrow KO_2 + H_2O$$

$$HgCl_2 + 2KI \longrightarrow HgI_2\downarrow + 2KCL$$
(scarlet Red)

IMP.

$$HgI_2 + 2KI \longrightarrow [K_2HgI_4] \longrightarrow \text{(soluble)}$$

Alkaline K_2HgI_4 is known as Nessler's reagent. It is used for the detection of NH_4^+ ion.



⟶ Brown colour
(Iodide of Million's base)

GROUP - IIA

REDUCING NATURE

Ba > Sr > Ca > Mg > Be

(BP, MP) :- Be > Mg < Ca > Sr > Ba

Hydration energy - Be > Mg > Ca > Sr > Ba

All IIA elements dissolve in mercury - Amalgame

Density - Be < Mg > Ca < Sr < Ba (calcium is least)

No superoxide is formed. only (Sr, Ba) form perosxides (SrO_2, BaO_2).

All IIA react with N_2 to form metal nitride.

$$3Ca + N_2 + Ca_3N_2 \longrightarrow Mg_3N_2, 6H_2O + 2NH_3\uparrow + 3Mg(OH)_2$$

(Be, Mg) → Covalent hydrides; (Ca, Sr, Ba) → Ionic hydrides

$Ca + H_2 \longrightarrow CaH_2$ (Hydrolith)

WITH NH_3

$$M + (x+y)NH_3 \longrightarrow \underset{\text{(solvated metal)}}{M^{+2}(NH_3)_x} + \underset{\text{(solvated e}^-\text{)}}{2e^-(NH_3)y}$$

Deep blue solution (highly conductive)

$$Ca + 2NH_3 \longrightarrow \underset{\text{(Ionic)}}{Ca(NH_2)_2} + H_2\uparrow$$

WITH H_2O

$$Be + 2H_2O \xrightarrow{\text{steam}} Be(OH)_2 + H_2\uparrow$$ (steam has high heat of vapourisn)

$$Mg + 2H_2O \xrightarrow{\text{(hot)}} Mg(OH)_2 + H_2\uparrow$$

$$Ca + 2H_2O \xrightarrow{\text{(hot)}} Ca(OH)_2 + H_2\uparrow$$

$$(Ba, Sr) + H_2O \xrightarrow{\text{(cold)}} (Ba(OH)_2, Sr(OH)_2) + H_2\uparrow$$

Reactivity with H_2O ↑ down the group.

IONIC CARBIDES

Methanides

Be_2C, Al_4C_3 $\xrightarrow{H_2O}$ CH_4

Allylides

Mg_2C_3 $\xrightarrow{H_2O}$ $CH_3-C\equiv CH$

Acetylides

Li_2C_2 BaC_2 CaC_2 SrC_2

$\xrightarrow{H_2O}$ $CH\equiv CH$

Solubility – $BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$

Stability – $BeSO_4 < MgSO_4 < CaSO_4 < SrSO_4 < BaSO_4$

HEATING

$$CaSO_4 \xrightarrow{\Delta} CaO + SO_3$$

$$MgSO_4 \cdot 7H_2O \xrightarrow[<160^\circ C]{\Delta} MgSO_4 \cdot H_2O + 6H_2O \xrightarrow{160^\circ C} MgSO_4 + H_2O \xrightarrow{200^\circ C} MgO + SO_2 + SO_3 + O_2$$

(Epsom)

$$Ca(OH)_2 + CO_2 \longrightarrow \underset{\text{(white)}}{CaCO_3} + H_2O \xrightarrow[\text{(excess)}]{CO_2} \underset{\text{(sol. in } H_2O)}{Ca(HCO_3)_2}$$

All phosphate are insoluble in H_2O except IA. (one)

Stability – $BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$

Solubility – $BeCO_3 > MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$

SUPER PHOSPHATES

$Ca_3(PO_4)_2$ [Rock Phosphate] $\longrightarrow$ Insoluble

$+$

$2H_2SO_4 \longrightarrow CaSO_4 + CaH_2(PO_4)_2 \longrightarrow \underset{\text{Insoluble}}{2CaSO_4} + \underset{\text{Soluble}}{Ca(H_2PO_4)_2}$

Calcium superphosphate $\longrightarrow$ Fertiliser

$Ca_3(PO_4)_2 + 4H_3PO_4 \longrightarrow 3Ca(H_2PO_4)_2$

Calcium triple superphosphate $\longrightarrow$ Better Fertiliser

Solubility :- $Ca(H_2PO_4) > CaHPO_4 > Ca_3(PO_4)_2$

Insoluble

Baryta water $\longrightarrow Ba(OH)_2$

$$Ba(OH)_2 + CO_2 \longrightarrow BaCO_3\downarrow + H_2O$$

BLEACHING POWDER

$3Ca(OH)_2 + 2Cl_2 \longrightarrow Ca(OH)_2 \cdot Ca(OCl)_2 \cdot CaCl_2 \cdot 2H_2O$

($3Ca(OH)_2$: slaked lime)

Active component of compound is $Ca(OCl)_2$

$Ca(Cl)(Cl) \longrightarrow$ calcium hypo chlorite ; Formula $\longrightarrow CaOCl_2$

GYPSUM

$CaSO_4 \cdot 2H_2O$

$CaSO_4 \cdot 2H_2O \xrightarrow[\text{(Hard)}]{120^\circ C} CaSO_4 \cdot \frac{1}{2}H_2O \xrightarrow[\text{P.O.P}]{200^\circ C} CaSO_4\ \frac{1}{2}H_2O \xrightarrow[\text{Red burnt Plaster}]{500^\circ C} (CaO\ SO_3)$

CaO (QUICK LIME)

$Ca(OH)_2 \xrightarrow{\Delta} CaO + CO_2\uparrow$

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2\uparrow$

$Ca(NO_3)_2 \xrightarrow{\Delta} CaO + NO_2\uparrow + O_2\uparrow$

NH_3 gas is derived by using CaO.

CaO is used as a flux.

CaO (basic flux) $+ SiO_2 \longrightarrow CaSiO_3$

It is used as the lining of the furnace (refractory material)

$MgCO_3 \xrightarrow{\Delta} MgO + CO_2$

ANHYDRONE

It is dehydrating agent.

$Mg(ClO_4)_2 \xrightarrow[\text{(Absorb)}]{\Delta\ (200^\circ C)} Mg(ClO_4)_2 \cdot 6H_2O$

Sorel's cement – $MgCl_2 \cdot 5H_2O \cdot xH_2O$

$MgCl_2 \cdot 8H_2O$, $CaCl_2 \cdot 6H_2O$, $SrCl_2 \cdot 6H_2O$, $BaCl_2 \cdot 2H_2O$

GROUP - 13

PHYSICAL STATE

B, Al, Ga, In, Tl

B → Non-metal

Al, Ga, In, Tl → Metals

Ga → Exists in Liquid state during summers.

OXIDATION STATE

Boron can't form '+3' ion, it forms covalent compds.
Al forms '+3' ion.
Ga, In, Tl can both form '+1' and '+3'.
Down the group, the stability of the '+1' O.S increases due to the inert pair effect.

REACTIVITY TOWARDS AIR

$$2E + 3O_2(g) \xrightarrow{\Delta} 2E_2O_3(s)$$
$$2E + N_2(g) \xrightarrow{\Delta} 2EN(s)$$

Boron trioxides are acidic; Ga_2O_3 and Al_2O_3 are amphoteric; In_2O_3 and Tl_2O_3 are basic in nature.

REACTIVITY WITH ACIDS AND ALKALIS

(a) Boron doesn't react with acids and alkalis.

(b) Aluminium dissolves in both; showing amphoteric behaviours.

$$2Al(s) + 6HCl(aq) \longrightarrow 2AlCl_3(aq) + 3H_2(g)$$
(HCl: Acid)

$$2Al(s) + 2NaOH(aq) + 2H_2O(l) \longrightarrow 2Na[Al(OH)_4](aq) + 3H_2(g)$$
(NaOH: Base)

$Al(OH)_3$ primarily acts as a base.

(c) Tl_2O_3 and In_2O_3 are basic and do not form hydrates or hydroxides.

BORON (B)

(a) Occurance of Boron

i. Borax $[Na_2B_4O_7 \cdot 10H_2O]$
ii. Boric acid $[H_3BO_3]$
iii. Kernite $[Na_2B_4O_7 \cdot 4H_2O]$
iv. Colamanite $[Ca_2B_6O_{11} \cdot 5H_2O]$

(b) Extraction of Boron

i. $Na_2B_4O_7 + 2HCl + 5H_2O \longrightarrow 4H_3BO_3 + 2NaCl$

$2H_3BO_3 \xrightarrow{\Delta} B_2O_3 + 3H_2O$

$B_2O_3 + 3Na/Mg \xrightarrow{\Delta} 2B$

ii. From KBF_4:- $KBF_4 + 3K \xrightarrow{\Delta} 4KF + B$

iii. $aBX_3 + 3H_2O \xrightarrow[W]{\text{Red hot}} 2B + bHX \quad (X = Cl, Br)$

iv. Van Arkel method:- $2BI_3 \xrightarrow[W]{\text{Red hot}} 2B + 3I_2 \uparrow$

v. $B_2H_6 \xrightarrow{\Delta} 2B + 3H_2 \uparrow$

(c) Properties of Boron

It exist in 5 forms = 4 crystalline + 1 amorphous

i. $4B + 3O_2 \longrightarrow 2B_2O_3$; $2B + N_2 \longrightarrow 2BN$

ii. $BN + 3H_2O \longrightarrow H_3BO_3 + NH_3$

iii. $2B + 2NaOH + 2H_2O \longrightarrow NaBO_2 + 3H_2$

iv. $4B + 3SiO_2 \longrightarrow 2B_2O_3 + 3Si$

BORON TRIOXIDE (B_2O_3)

PREPARATION

$$H_3BO_3 \xrightarrow{100^\circ C} HBO_2 \xrightarrow{160^\circ C} H_4B_2O_7 \xrightarrow{\text{Red hot}} B_2O_3$$

PROPERTIES

Acidic oxide $\longrightarrow N_2O + B_2O_3 \longrightarrow Na_3BO_3$

It forms coloured compounds with transition metal salts.

$$B_2O_3 + Cr_2(SO_4)_3 \longrightarrow 2Cr(BO_3)_3 + SO_3\uparrow$$

$$B_2O_3 + Cu(NO_3)_2 \longrightarrow 2Cu(BO_2)_2 + NO_2\uparrow + O_2\uparrow$$

$$B_2O_3 + 6HF + 3H_2SO_4 \longrightarrow 2BF_3 + 3H_2SO_4 \cdot H_2O$$

ORTHOBORIC ACID (H_3BO_3)

PREPARATION

(i) $Na_2B_4O_7 + H_2SO_4 + 5H_2O \longrightarrow Na_2SO_4 + H_3BO_3\downarrow$

(ii) $Ca_2B_6O_{11} + 4SO_2 + 11H_2O \longrightarrow 2Ca(HSO_3)_2 + 6H_3BO_3\downarrow$

PROPERTIES

(i) H_3BO_3 is a weak monobasic acid.

(ii) $B(OH)_3 \quad 2H_2O \quad [B(OH)_4]^-_{(aq)} \quad H_3O^+_{(aq)}$

(iii) H_3BO_3 is a lewis acid and not a proton donor like others.

(iv) To increase the acidic properties of the boric acid; as diols are added to its aqueous sol^n

Glycerol, catechol, mannitol, salicyclic acid etc.

$$[B(OH)_4]^- + 2\begin{pmatrix} CH_2OH \\ CH_2OH \end{pmatrix} \rightleftharpoons \left[\begin{array}{c} O \\ O \end{array}\!\!>\!B\!<\!\!\begin{array}{c} O \\ O \end{array}\right] + H^+ + 4H_2O$$

(v) When heated; it forms B_2O_3.

$$H_3BO_3 \xrightarrow{100^\circ C} HBO_2 \xrightarrow{160^\circ C} H_2B_4O_7 \xrightarrow[hot]{red} B_2O_3$$

(vi) H_3BO_3 is greasy to touch; less soluble in cold water; but more soluble in hot water. In solid state, the $B(OH)_3$ units are hydrogen bonded in 2-D sheets.

(vii) Test of boric acid is done using ethanol

$$H_3BO_3 + 3EtOH \longrightarrow (EtO)_3B + 3H_2O$$

(viii) Polymeric metaborates are formed at high conc.^{n}

$$3B(OH)_3 \rightleftharpoons H_3O^+ + [B_2O_3(OH)_4] + H_2O$$

BORAX ($Na_2B_4O_7 \cdot 10H_2O$)

PREPARATION

(i) $Ca_2B_6O_{11} + Na_2CO_3 \longrightarrow Na_2B_4O_7 + NaBO_2 + CaCO_3$

(ii) $4H_3BO_3 + Na_2CO_3 \longrightarrow Na_2B_4O_7 + 6H_2O + CO_2\uparrow$

PROPERTIES

(i) It's aqueous solution is alkaline because of hydrolysis to weak acid (H_3BO_3) and strong base (NaOH)

$$Na_2B_4O_7 + 7H_2O \longrightarrow 4H_3BO_3 + 2NaOH$$

$$Na_2B_4O_7 \cdot 10H_2O \xrightarrow{\Delta} Na_2B_4O_7 + 10H_2O\uparrow$$

$$Na_2B_4O_7 \xrightarrow{740^\circ C} 2NaBO_2 + B_2O_3 \text{ (Borax - bead)}$$

Oxidation with H_2O_2

$$Na_2B_4O_7 \xrightarrow{\Delta} NaBO_2 + 2H_2O_2 + 6H_2O \longrightarrow Na_2[(OH)_2B(O-O)_2B(OH)_2].6H_2O$$

used as a brightener in the washing powder — sodium peroxoborate

one mole of borax reacts with 2 moles acid.

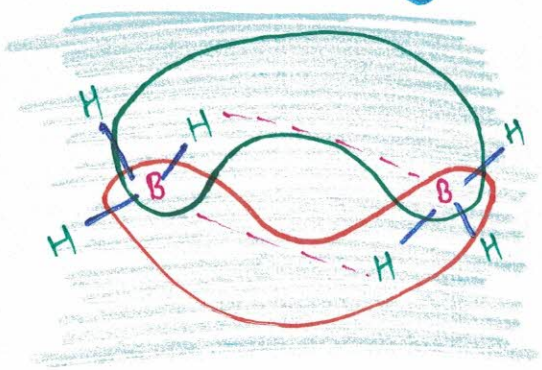
$$[B_4O_5(OH)_4]^{-2} + 5H_2O \rightleftharpoons 2H_3BO_3 + 2[B(OH)_4]^-$$

H_3BO_3 is a weak acid, hence only $[B(OH)_4]^-$ will react with an acid.

$$2[B(OH)_4] + H_3O^+ \rightleftharpoons 2B(OH)_3 + 4H_2O$$

Borax is also used as a buffer.
Correct formula of borax is $Na_2[B_4O_5(OH)_4].8H_2O$
It has '5' B-O-B bonds and '8' water of crystallization.

DIBORANE (B_2H_6)



It has 3 centered 2e⁻ bond; banana bonds, the B-H bridge bond is longer and stronger than the terminal B-H bonds.

PREPARATION

(i) $BF_3 + LiAlH_4 \xrightarrow{\text{ether}} 2B_2H_6$

(ii) $BCl_3 + 6H_2(\text{excess}) \xrightarrow[\text{discharge}]{\text{silent electric}} B_2H_6$

(iii) $2BF_3 + 6NaOH \longrightarrow B_2H_6 + 6NaF$

PROPERTIES

(i) B_2H_6 is a colourless and reactive gas.

(ii) It has a high colorific value.

$$B_2H_6 + 3O_2 \longrightarrow B_2O_3 + 3H_2O \qquad \Delta H < 0$$

(iii) $B_2H_6 + 6H_2O \longrightarrow 2B(OH)_3 + 6H_2$

(iv) $B_2H_6 + 6Cl_2 \longrightarrow 2BCl_3 + 6HCl$

REACTION WITH NH_3 and amines

(i) Small amines (NH_3, 1° amine, 2° amine) give the unsymmetrical cleavage.

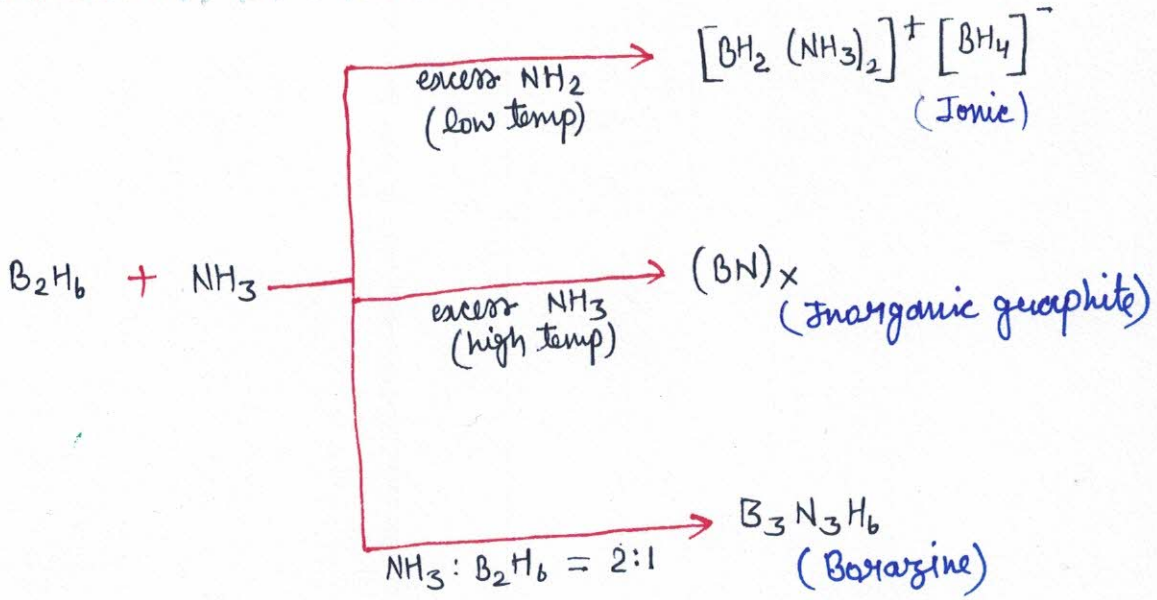
$$B_2H_6 + 2NH_3 \longrightarrow [H_2B(NH_3)_2]^+[BH_4]$$

(ii) Larger amines (3° amines, pyridine) give the symmetrical cleavage.

$$2(CH_3)_3N + B_2H_6 \longrightarrow 2H_3B \leftarrow N(CH_3)_3$$

Adduct

Reaction of B_2H_6 with NH_3 in different condition-



ALUMINIUM

Al_2O_3 (Aluminium oxide) – Alumina
It occurs in the form of bauxite.

PREPARATION

$$2Al(OH)_3 \xrightarrow{\Delta} Al_2O_3 + 3H_2O\uparrow$$

- By aluminium chloride ($Al_2Cl_3 \cdot 6H_2O$)

$$2Al + 3HCl \longrightarrow 2AlCl_3 + 3H_2\uparrow$$
$$Al_2O_3 + 6HCl \longrightarrow 2AlCl_3 + 3H_2O\uparrow$$
$$Al_2O_3 + 3C + 3Cl_2 \longrightarrow 2AlCl_3 \text{ (anhyd.)} + 3CO\uparrow$$

PROPERTIES

Aqueous solution is acidic in nature.

$$[Al(H_2O)_6]^{+3} \rightleftharpoons [Al(H_2O)_5OH]^{+2} + H^+$$

ALUMS

(a) $M_2SO_4 \cdot M_2(SO_4)_3 \cdot 24H_2O$
where M is any cation (univalent ; except Li^+)
Where M is trivalent cation (eg. – Al^{+3}, Cr^{+3} Fe^{+3} Co^{+3} etc)

(b) Alums are double salts which when dissolved in water produce metal and sulphate ions

These are used as –

(i) Mordant in dye industry.
(ii) Germicide for water purification.
(iii) coagulating agent.

GROUP – 14

CARBON FAMILY

MEMBER — C, Si, Ge, Sn, Pb

Anamalous behaviour of carbon – Due to the high E.N., small size, high I.E, no d-orbitals. It shows catenation properly

Allotrops of Carbon –

Diamond – Crystalline lattice, sp^3 hybridisation, 3D network of carbon atoms, directional covalent bonds are present, hardest natural substance on earth

Graphite – Layered structure, vanderwall's force between the two layers, sp^2 hybridisation, one free e^- thermodynamically more stable than diamond, good conductor of electricity.

Fullerene – Smooth structure, 20 six membered rings, 12 five membered rings, sp^2 hybridisation

E.g C_{60} molecule

CARBON MONOXIDE (CO)

PREPARATION

(i) $C_{(s)} + CO_{2(g)} \longrightarrow 2CO_{(g)}$





(iv) $C_{(s)} + H_2O_{(g)} \longrightarrow CO_{(g)} + H_{2(g)}$ (water gas)

$C_{(s)} + O_{2(g)} + 4N_{2(g)} \longrightarrow 2CO + 4N_{2(g)}$ (Producer gas)

$K_4[Fe(CN)_6] + 6H_2SO_4 + 6H_2O \longrightarrow K_2SO_4 + FeSO_4 + 6CO + 3(NH_4)_2SO_4$

PROPERTIES

(i) Colourless odourless gas; burns with a blue flame, neutral oxide, toxic, forms carboxyhaemoglobin with blood which can lead its death.

(ii) CO is a powerful reducing agent in the extraction of Fe and Ni.

$Fe_2O_{3(s)} + CO_{(g)} \longrightarrow 2Fe_{(s)} + 2CO_{2(g)}$

$NiO_{(s)} + CO_{(g)} \longrightarrow Ni_{(s)} + CO_{2(g)}$

(iii) Mond's process –

$Ni_{(s)} + 4CO_{(g)} \xrightarrow{28^\circ C} Ni(CO)_4 \xrightarrow{180^\circ C} Ni + 4CO$

(iv) $CO + Cl_2 \longrightarrow COCl_2$ (Phosagene)

CARBON DIOXIDE (CO_2)

$CO_3^{-2}{}_{(aq)} + 2H^+_{(aq)} \longrightarrow CO_{2(g)} + H_2O_{(l)}$

(i) colourless odourless gas; used in fire extinguishers.

(ii) H_2CO_3/HCO_3^- is a buffer solution

(iii) Lime water Test –

$Ca(OH)_2 + CO_2 \longrightarrow CaCO_{3(s)} + H_2O$

$CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_{2(aq)}$

CARBON SUBOXIDE (C_3O_2)

O=C=C=C=O

(i) $CH_2(COOH) + P_4O_{10} \longrightarrow 3C_3O_2 + 4H_5PO_4$
(Malonic acid)

(ii) $C_3O_2 \xrightarrow{\Delta} CO_2 + 2C$

CARBIDES

@. IONIC CARBIDES

(i) Methanides - C^{4-}

It gives methane on hydrolysis.

$$Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$$

$$Be_2C + 4H_2O \longrightarrow 2Be(OH)_2 + CH_4$$

(ii) Acetylides - C_2^{-2}

It gives acetylene on hydrolysis.

$$CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + C_2H_2$$

$$Al_2(C_2)_3 + 6H_2O \longrightarrow 2Al(OH)_3 + 3C_2H_2$$

(iii) Allylides - C_3^{-4}

It gives propyne on hydrolysis.

$$Mg_2C_3 + 4H_2O \longrightarrow 2Mg(OH)_2 + CH_3-C\equiv CH$$

b. COVALENT CARBIDES

CH_4, CO_2 CS_2 SiC B_4C etc.

c. INTERSTICIAL CARBIDES

These are formed by transition metals; carbon atoms occupy octahedral voids, very hard in nature; they have high M.P, conduct electricity.

E.g. $-N^{\oplus}\equiv C^{\ominus}$

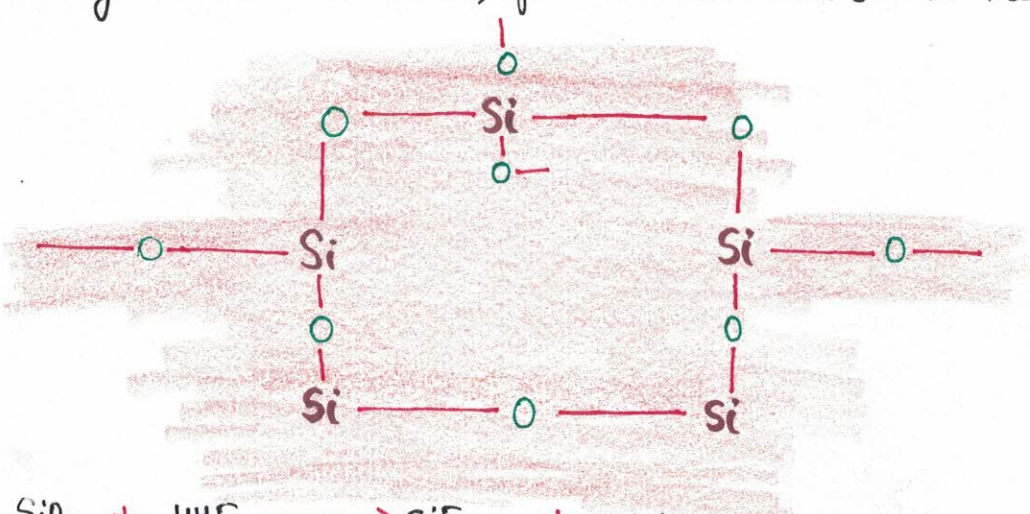
SILICON (Si)

$SiO_2 + 2Mg \longrightarrow 2MgO + Si$

SILICON DIOXIDE (SiO_2)

(a) Silica is a covalent, 3D network solid.

(b) Each Si atom is tetrahedrally bonded to the four 'O' atoms; each corner is shared by another tetrahedron; giant molecule with 8-membered rings.



(c) $SiO_2 + 4HF \longrightarrow SiF_4 + 2H_2O$

SILICATES

Co-ordination no. of 'Si' is four; sp^3 hyb.

(a) **Orthosilicates** - Discrete $[SiO_4]^{-4}$ units; no sharing of corners with other atoms.

E.g. $Zr SiO_4$ $Mg SiO_4$ $Zn SiO_4$

(b) **Pyrosilicates** - $[SiO_7]^{-6}$ units; 2 tetrahedral units share 'one' corner oxygen atom.

E.g. $Sc_2 Si_2O_7$

(c) **Cyclic Silicates** - $[SiO_3]^{2n-}$ unit, two oxygen atoms per tetrahedron are shared.

E.g. Emerald

(d) Chain silicates

- Pyroxenes – Simple chain $(SiO_3)_n^{2n-}$
- Amphibales – Double chain $(Si_4O_{11})_n^{6n-}$

(e) 2-D sheet silicates

– 3 oxygen atoms are shared.
SiO_4^{4-} unit

Trick · Decrease the oxygen atom by 0.5 and charge by 1 on going bottom

Remember Amphibales gen. formula – $(Si_4O_{11})^{6-}$

SILICATE	No. of 'O' atoms shared per unit	Gen. Formula
Ortho	0	SiO_4^{4-}
Pyro	1	$(SiO_{3.5})^{-3}$
cyclic / Single chain	2	$(SiO_3)^{-2}$
2-D	3	$(SiO_{2.5})^{-}$
3-D	4	SiO_2

SILICONES

(i) Synthetic organosilicon compounds having R_2SiO units held by Si–O–Si linkages

(ii) General formula :- $[R_2(SiO)]_n$

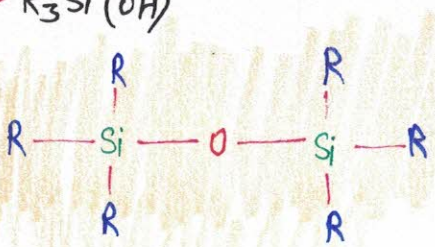
(iii) Silicons are formed by alkyl and aryl structures (substituted chlorosilanes) and their subsequent polymerisation.

$$RCl + Si \xrightarrow[300^\circ C]{Cu} R_3SiCl + R_2SiCl_2 + RSiCl_3$$

(a) Silicones from R_3SiCl

$$R_3SiCl + H_2O \longrightarrow R_3Si(OH)$$

$$2R_3SiOH \xrightarrow{\text{condensation}}$$

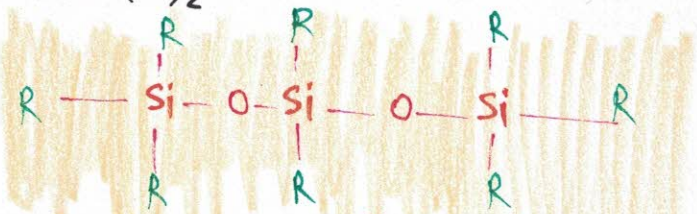


b. Silicones from R_2SiCl_2

$R_2SiCl_2 \quad 2H_2O$

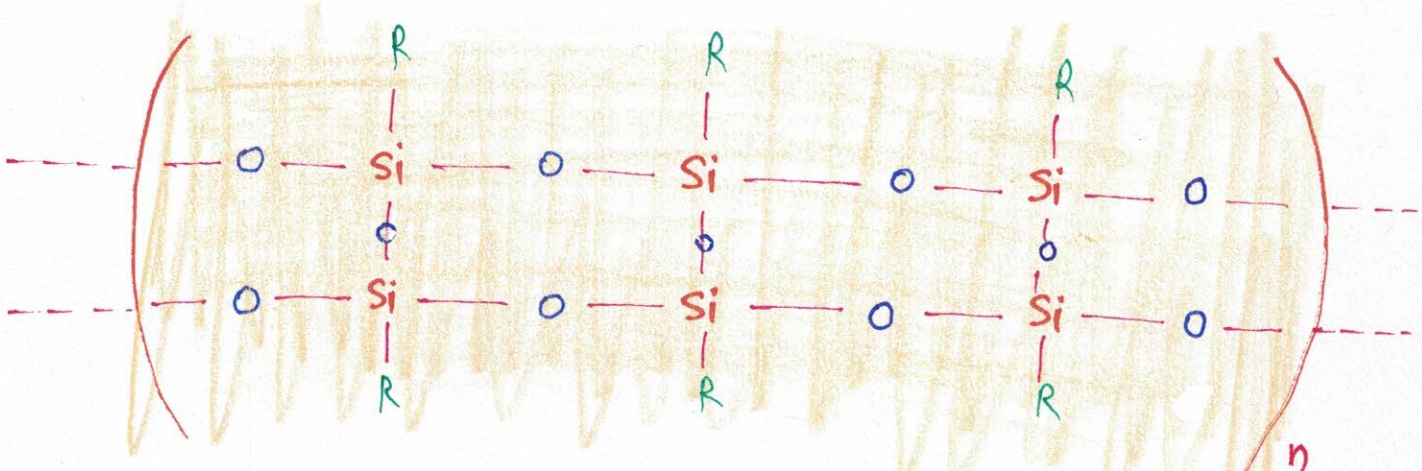
$R_2Si(OH)_2$

$R_2Si(OH)_2$



(Long chain polymer)

c. Silicones from $RSiCl_3$



Cross-linked polymer

LITHARGE (PbO)

$2Pb + O_2 \xrightarrow{\Delta} 2PbO$ (Amphoteric oxide)

a. Lead Oxide (PbO_2)

(i) $PbO + NaOCl \longrightarrow PbO_2 + NaCl$

(ii) $Pb_3O_4 + 4HNO_3\,(dil.) \longrightarrow 2Pb(NO_3)_2 + PbO_2 + 2H_2O$

(PbO_2: chocolate brown solid)

(iii) $2PbO_2 \xrightarrow{\Delta} 2PbO + O_2$

(iv) PbO_2 is an oxidising agent. ($Pb^{+4} > Pb^{+2}$) $\longrightarrow$ O.A

(v) $PbO_2 + 2HNO_3 \longrightarrow Pb(NO_3)_2 + \frac{1}{2}O_2 + H_2O$

Red lead (Pb_3O_4)

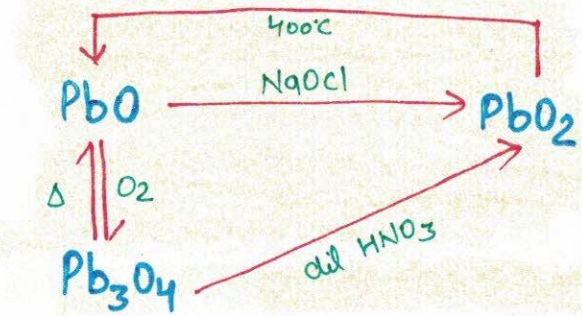
(i) $6PbO + O_2 \xrightarrow{\Delta} 2Pb_3O_4$

(ii) $Pb_3O_4 + 4HNO_3 \longrightarrow 2Pb(NO_3)_2 + PbO_2 + 6H_2O$

(iii) $Pb_3O_4 \longrightarrow 6\,PbO \quad O_2$

(iv) $Pb_3O_4 + 8HCL \longrightarrow 3PbCl_2 + 4H_2O + Cl_2$

(v) $2Pb_3O_4 + 6H_2SO_4 \longrightarrow 6PbSO_4 + 6H_2O + O_2$



PbO_2, Pb_3O_4 $\longrightarrow PbSO_4 + O_2 \uparrow$

$Pb_3O_4 \longrightarrow PbCl_2 + Cl_2 \uparrow$

$PbO_2 \longrightarrow Pb(NO_3)_2 + O_2 \uparrow$

$Pb_3O_4 \longrightarrow$ No oxygen

GROUP - 15

NITROGEN FAMILY

Members – N, P, As, Sb, Bi

Oxidation state – Common O.S – {-3, +3, +5}

Stability – $Bi^{+3} > Sb^{+3} > As^{+3}$; $As^{+5} > Sb^{+5} > Bi^{+5}$

"N" exhibits {+1, +2, +4} O.S when it reacts with oxygen.

In "N" all O.S from "+1" to "+4" tend to disproportionate in acidic medium.

$$3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$$

Physical state –

- N, P – Non-metals
- As, Sb – Metalloids
- Bi – Metals

DINITROGEN (N_2)

PREPARATION

(i) $NH_4Cl + NaNO_2 \longrightarrow N_2\uparrow + H_2O + NaCl$

(ii) $NH_4NO_2 \xrightarrow{\Delta} N_2\uparrow + 2H_2O$

(iii) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2\uparrow + 4H_2O + Cr_2O_3$

(iv) $Ba(N_3)_2 \xrightarrow{\Delta} Ba\uparrow + 3N_2$

(v) By Liquefaction of air.

PROPERTIES

N_2 is a colourless, odourless gas insoluble in H_2O.

Metals reacts with N_2 to form nitrides.

$$Mg + N_2 \longrightarrow Mg_3N_2 + H_2O \longrightarrow Mg(OH)_2 + NH_3$$

$$N_2 + 3H_2 \longrightarrow 2NH_3 \quad \text{(Born Haber's Process)}$$

(iv) $N_2 + O_2 \longrightarrow 2NO$

(v) $CaC_2 + N_2 \longrightarrow CaCN_2 + C$ (Nitralium Fertiliser)

$BaC_2 + N_2 \longrightarrow Ba(CN)_2$

AMMONIA (NH_3)

PREPARATION

(i) $NH_4NO_3 + NaOH \longrightarrow NaNO_3 + H_2O + NH_3$

(ii) $AlN + NaOH + H_2O \longrightarrow NaAlO_2 + NH_3\uparrow$

(iii) Ammonia can't be dried using $CaCl_2$, P_2O_5 or conc. H_2SO_4. It is dried by CaO.

$CaCl_2 + 8NH_3 \longrightarrow CaCl_2 \cdot 8NH_3$ | $P_2O_5 + 6NH_3 + 3H_2O \longrightarrow 2(NH_3)_3PO_4$

$H_2SO_4 + 2NH_3 \longrightarrow (NH_4)_2SO_4$ | $CaO + NH_3 + NH_3 \longrightarrow XX$

(iv) $H_2N-\overset{\overset{O}{\|}}{C}-NH_2 + 2H_2O \longrightarrow (NH_4)_2CO_3 \rightleftharpoons 2NH_3 + H_2O + CO_2$

(v) Haber's Process – $N_2 + 3H_2 \xrightarrow[Fe_2O_3 + K_2O/Al_2O_3]{500^\circ C,\ 200\ atm} 2NH_3\uparrow$

(vi) Cyanamide Process – $CaO + 2C + N_2 \longrightarrow CaCN_2 + CO\uparrow$

$CaCN_2 + 3H_2O \longrightarrow CaCO_3 + 2NH_3\uparrow$

(vii) From destructive distillation of coal.

PROPERTIES

(i) If ammonia gas is evolved by a compound then we can only say that it is a nitrate or a nitrite.

(ii) NH_3 is a colourless gas with a pungent odour. In solid and Liquid state, it undergoes H-bonding due to which it has high boiling point than expected.

(iii) It's aqueous solution is weakly basic in nature.

$NH_3 + H_2O\ (l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

(iv) (a) $FeCl_3 + NH_4OH \longrightarrow Fe_2O_3.xH_2O$ (brown ppt.) $+ NH_4Cl$

(b) $ZnSO_4 + 2NH_4OH \longrightarrow Zn(OH)_2$ (white ppt) $+ (NH_4)_2SO_4$

(v) It acts as a ligand in complex compds.

(a) Cu^{+2} (blue) $+ 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{+2}$ (deep blue)

(b) Ag^+ (colourless) $+ Cl^- \rightleftharpoons AgCl$ (white ppt)

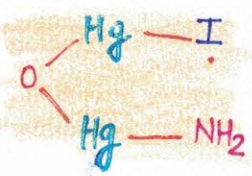
$AgCl + 2NH_3(aq) \longrightarrow [Ag(NH_3)_2Cl]$ (colourless)

(vi) If the anion part in the ammonium salt is not a strong oxidising agent, then on heating it gives NH_3.

$NH_4Cl \xrightarrow{\Delta} NH_3 + HCl$ (Less oxidising)

$NH_4NO_2 \xrightarrow{\Delta} N_2 \quad 2H_2O$ (More oxidising)

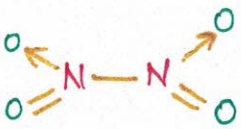
(vii) Tests of Ammonia – using Nessler's reagent.

$NH_3 + 2K_2HgI_4 + 3KOH \longrightarrow$  $+ 7KI + 2H_2O$

OXIDES OF NITROGEN

N_2O 

N_2O_3 

N_2O_4 

NO 

NO_2 

N_2O_5 

OXIDE	PHYSICAL NATURE	PREPARATION
N_2O	Colourless gas, Neutral	$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$
NO	Colourless gas, Neutral	$2NaNO_2 + 2FeSO_4 + 3H_2SO_4 \longrightarrow Fe(SO_4)_3 + 2NaHSO_4 + 2H_2O + 2NO$
N_2O_3	Blue Liquid, acidic	$2NO + N_2O_4 \xrightarrow{250K} 2N_2O_3$

NO_2 — Brown gas, acidic

$2Pb(NO_3)_2 \longrightarrow 4NO_2 + 2PbO.O_2$

N_2O_4 — Colourless solid, acidic

$2NO_2 \underset{\text{Heat}}{\overset{\text{cool}}{\rightleftharpoons}} N_2O_4$

N_2O_5 — Colourless solid acidic

$4HNO_3 + P_4O_{10} \longrightarrow 4HPO_3 + N_2O_5$

OXYACIDS OF NITROGEN

Nitrous acid (HNO_2)

Preparation –

(a) $Ba(NO_2)_2 + H_2SO_4 \longrightarrow 2HNO_2 + BaSO_4 \downarrow$

(b) $3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$

Nitric acid (HNO_3)

Preparation –

(a) $NaNO_3 + H_2SO_4 \longrightarrow NaHSO_4 + HNO_3$

(b) Ostwald's process – $4NH_3 + 5O_2 \xrightarrow[500K,\ 9bar]{Pt/Rh\ gauge} 4NO + 6H_2O(g)$

$2NO + O_2 \rightleftharpoons 2NO_2(g)$

$3NO_2 + H_2O \longrightarrow 2HNO_3 + NO(g)$

(can be used again)

ii) -ve SRP metal → More reactive

- Very dil. HNO_3 → Ammonium nitrate or ammonic + MNO_3
- dil. HNO_3 → $MNO_3 + NO$
- Conc. HNO_3 → $MNO_3 + NO_2$

(ii) +ve SRP metal / Less reactive

- very dil. HNO_3 → X
- dil. HNO_3 → $MNO_3 + N_2O$
- conc. HNO_3 → $MNO_3 + NO_2$

(iii) Non Metal / Method → "ic acid" + NO_2

BROWN RING TEST

Given by nitrates

$$NO_3^- + 3Fe^{+2} + 4H^+ \longrightarrow NO + 3Fe^{+3} + 2H_2O$$

$$[Fe(H_2O)_6]^{+2} + NO \longrightarrow [Fe(H_2O)_5(NO)]^{+2} + H_2O$$

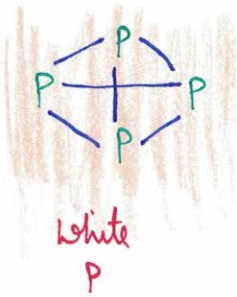
In brown ring, Fe is in +1 O.S and NO^+ acts as a ligand.

ALLOTROPES OF PHOSPHORUS

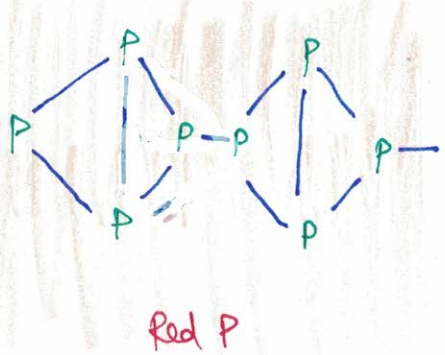
White P_4 (Insoluble in H_2O, soluble in CS_2) —High Pressure, 473K→ Black P_4 (β)

White P_4 —Δ (Inert atm. of CO_2), 573K→ Red P_4 (Insoluble in H_2O and CS_2) —803K (In a sealed tube)→ Black P_4 (α) (Monoclinic or rhombohedral structure)

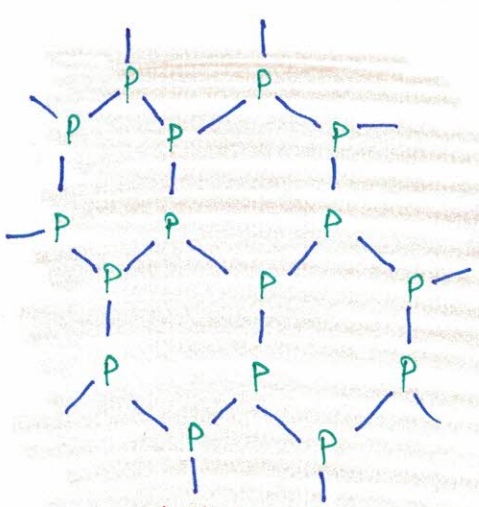
REACTIVITY ORDER – White > Red > Black



White P



Red P



Black P

PHOSPHINE (PH_3)

PREPARATION

(i) $Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$

(ii) White $P_4 + 3NaOH + 3H_2O \longrightarrow PH_3 + 3NaH_2PO_2$

(iii) $PHI + KOH \longrightarrow KI + H_2O + PH_3$

(Phosphonium Iodide)

PROPERTIES

(i) PH_3 is a colourless gas with rotten fish smell and is highly poisonous.

(ii) PH_3 is slightly soluble in H_2O but is insoluble in CS_2 and other organic solvents.

(iii) $PH_3 + HBr \longrightarrow PH_4Br$ (Phosphonium bromide)

(iv) $3\,Cu_{SO_4} + PH_3 \longrightarrow Cu_3P_2 \downarrow + 3H_2SO_4$

$3HgCl_2 + 2PH_3 \longrightarrow Hg_3P_2 \downarrow + 6HCl$

a. $P_4 + 6Cl_2 \longrightarrow 4PCl_3$

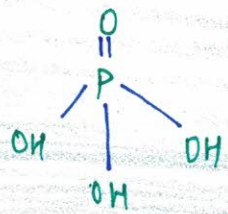
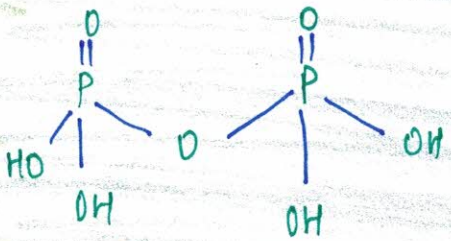
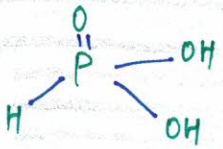
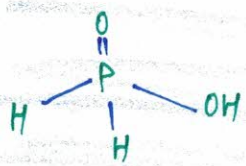
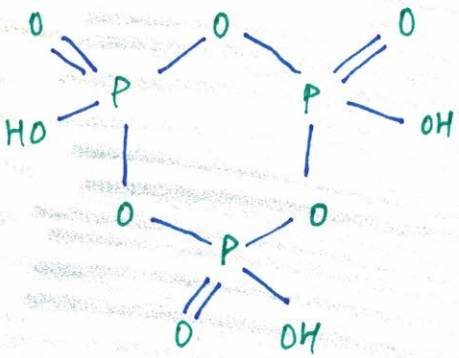
b. $P_4 + 10Cl_2 \longrightarrow 4PCl_5$

c. $P_4 + 8SOCl_2 \longrightarrow 4PCl_3 + 4SO_2 + 2S_2Cl_2$

$P_4 + 10SOCl_2 \longrightarrow 4PCl_5 + 10SO_2$

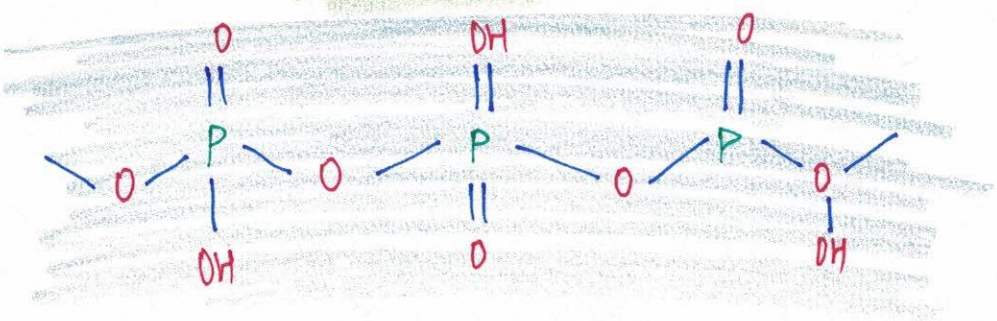
OXOACIDS OF PHOSPHORUS

NAME	FORMULA	PREPARATION OF COMPOUND
Hypophosphorous (Phosphinic)	H_3PO_2 (+1)	White P_4 + alkali
Orthophosphorous (Phosphonic)	H_3PO_3 (+3)	$P_2O_3 + H_2O$
Pyrophosphorous	$H_4P_2O_5$ (+3)	$PCl_3 + H_3PO_3$

NAME	FORMULA	PREPARATION OF COMPOUND
Hypophosphoric	$H_4P_2O_6$ (+4)	Red P_4 + Alkali
Orthophosphoric	H_3PO_4 (+5)	$P_4O_{10} + H_2O$
Pyrophosphoric	$H_4P_2O_7$ (+5)	Heat phosphoric acid
Metaphosphoric	$(HPO_3)_n$ (+5)	Phosphorous acid + Br_2, heat in sealed tube.
Orthophosphoric acid	H_3PO_4	
Pyrophosphoric acid	$H_4P_2O_7$	
Orthophosphorous acid	H_3PO_3	
Hypophosphorous acid	H_3PO_2	
cyclotrimetaphosphoric acid	$(HPO_3)_3$	

Poly meta phosphoric acid

$(HPO_3)_n$



$$4H_3PO_3 \longrightarrow 3H_3PO_4 + PH_3$$

$$4AgNO_3 + 2H_2O + H_3PO_2 \longrightarrow 4Ag + 4HNO_3 + H_3PO_4$$

GROUP - 16

OXYGEN FAMILY

Members - O S Se Te Po

OXIDATION STATE = oxygen mainly shows (+2) O.S except in OF_2 (+2).
Stability of +4 O.S down the group increases than +6.

DIOXYGEN (O_2)

PREPARATION

(i) $2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$

(ii) $2HgO \xrightarrow{\Delta} 2Hg + O_2$

$3MnO_2 \xrightarrow{\Delta} Mn_3O_4 + O_2$

$2Ag_2O \xrightarrow{\Delta} 4Ag + O_2$

$2Pb_3O_4 \xrightarrow{\Delta} 6PbO + O$

(iii) $4K_2Cr_2O_7 \xrightarrow{\Delta} 4K_2CrO_4 + 2Cr_2O_3 + 3O_2$

$2KMnO_4 \xrightarrow{\Delta} K_2Mn_2O_4 + MnO_2 + O_2$

OXIDES - Acidic oxides - CO_2 SO_2 SO_3 N_2O_5 P_4O_{10} ClO_7 etc, Mn_2O_7, CrO_3 etc.

Basic oxides - gen. metallic oxides, Na_2O, CaO, CrO, CuO, FeO, BaO

Neutral oxides - CO, N_2O, NO

Amphoteric oxides - ZnO, Al_2O_3, BeO, Sb_2O_3 Cr_2O_3 PbO etc

Mixed oxides - Pb_3O_4 ($2PbO + PbO_2$), Fe_3O_4 ($FeO + Fe_2O_3$), Mn_3O_4 ($2MnO + MnO_2$)

Peroxides - Na_2O_2, K_2O_2, BaO_2

Superoxides - KO_2, RbO_2 CsO_2

OZONE (O_3)

$3O_2 \rightleftharpoons 2O_3$ (silent electric discharge)

Dark blue gas, blue liquid, violet black solid.

Absorbs UV Light protects us.

O_3 is diamagnetic toxic, slightly soluble in H_2O.

Oxidising Properties -

$O_3 + 2H^+ + 2e^- \longrightarrow O_2 + H_2O$ (acidic medium)

$O_3 + H_2O + 2e^- \longrightarrow O_2 + 2OH^-$ (basic medium)

O_3 act as better oxidising agent in acidic medium.

$I^- \xrightarrow{O_3} I_2$

$S^{2-} \xrightarrow{O_3} SO^{2-}$

$NO_2^- \xrightarrow{O_3} NO_3^-$

$SO_3^{2-} \xrightarrow{O_3} SO_4^{2-}$

$AsO_3^{3-} \xrightarrow{O_3} AsO_4^{3-}$

$K_2MnO_4 \xrightarrow{O_3} KMnO_4$

$S \xrightarrow{O_3} H_2SO_4$

$H_2S \xrightarrow{O_3} S\downarrow$ (yellow)

$NO_2 \xrightarrow{O_3} N_2O_5$

$[Fe(CN)_6]^{4-} \xrightarrow{O_3} Fe(CN)_6$

Dry $I_2 + O_3 \longrightarrow I_4O_9 + 9O_2$ (Yellow ionic solid) $\longrightarrow I^{+3}(IO_3^-)_3$

Moist $I_2 + O_3 \longrightarrow I_2O_5 \xrightarrow{H_2O} HIO_3$

$Ag \xrightarrow{O_3} Ag_2O\downarrow$ (Black)

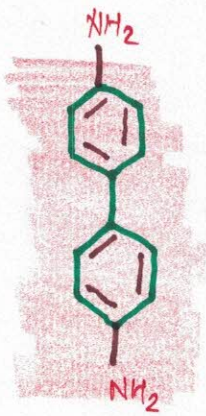
$H_2O \xrightarrow{O_3} H_2O + 2O_2$

$2KOH + 5O_3 \longrightarrow 2KO_3 + 5O_2 + H_2O$ (orang acid)

TEST FOR OZONE

- A filter paper soaked in alcholic benzidine becomes brown when brough in contact with.
- Tailing of Mercury - $2Hg + O_3 \longrightarrow H_2O + O_2$ (Hg_2O mercury suboxide)

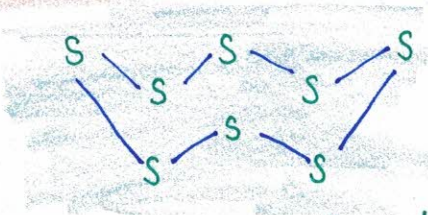
mobility of O_3 decreases due to formation of Hg_2O.



ALLOTROPIC FORMS OF SULPHUR

Rhombic Sulphur

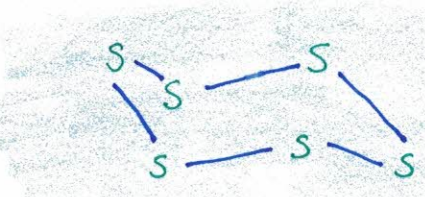
Yellow in colour. It's crystal are formed on evaporating the soln of roll sulphur in CS_2. It's insoluble in H_2O but soluble in CS_2.



Crown shaped structure S_8 ring.

Monoclinic Sulphur

It is soluble in CS_2, prepared by melting rhombic sulphur. It has colourless needle shaped structure.



$$\alpha \text{ sulphur (Rombic)} \underset{369\,K >}{\overset{369\,K <}{\rightleftharpoons}} \beta \text{ sulphur (Monoclinic)}$$

cyclo S_6 contains 6 - 20 sulphur atoms per ring. it has chair form.

EXTRACTION OF SULPHUR

$$H_2S + O_2 \xrightarrow{\text{burn}} SO_2 + H_2S \xrightarrow[\text{connector}]{\text{catalytic}} 3S(g) + 2H_2O(g)$$

HYDROGEN SULPHIDE (H_2S)

PREPARATION

$$FeS + H_2SO_2 \longrightarrow FeSO_2 + H_2S$$

$$Sb_2S_3 + 6HCl \longrightarrow 2SbCl_3 + 3H_2S$$

PROPERTIES

Colourless gas with rotten fish smell.
It act as a reducing agent.

$$X_2 \xrightarrow{H_2S} HX \qquad H_2O_2 \xrightarrow{H_2S} S$$

$$SO_2 \xrightarrow{H_2S} S \qquad HNO_3 \xrightarrow{H_2S} S$$

$$MnO_4 \xrightarrow[OH^-]{H_2S} MnO_2 \qquad MnO_4 \xrightarrow[H^+]{H_2S} Mn^{+2}$$

Its aqueous soln is weak dibasic acid

$$H_2S \rightleftharpoons HS^- + H^+ \rightleftharpoons S^{2-} + 2H^+$$

Formation of Polysulphides

$$NH_4OH + H_2S \longrightarrow (NH_4)_2S + 2H_2O$$

$$(NH_4)_2S \xrightarrow[\text{(excess)}]{H_2S} (NH_4)S_{x+1} + xH_2$$

(Yellow ammonium sulphide)

TEST FOR H_2S

Turns acidified lead acetate paper to black.
Gives violet colouration with alkaline sodium nitroprussside sol^n

SULPHUR DIOXIDE

PREPARATION -

$$S + O_2 \text{ (in air)} \xrightarrow{\text{burn}} SO_2$$

$$CaSO_4 \text{ (gypsum)} + C \xrightarrow{\Delta\ 1000^\circ C} 2CaO + SO_2 + CO_2$$

PROPERTIES

colourless gas with irritating smell.

- acidic nature - $H_2O + SO_2 \longrightarrow H_2SO_3$
- addition Reasc^n - $SO_2 + Cl_2 \xrightarrow{hv} SO_2Cl_2$ (sulphuryl chloride)

$$SO_2 + O_2 \xrightarrow[\text{asbestos}]{\text{Platinised}} SO_3$$

$$PbO_2 + SO_2 \longrightarrow PbSO_4$$

can act as both oxidising and reducing agent.
Bleaching action - Reductive, temporary bleaching

$$SO_2 + 2H_2O \longrightarrow H_2SO_4 + 2H$$

coloured substance + [H] $\longrightarrow$ colourless

- $Fe_2(SO_4)_3 \longrightarrow Fe_2O_3 + SO_3$
- $6\,H_2SO_4 + P_4O_{10} \longrightarrow 6\,SO_3 + 4H_3PO_4$
- $H_2SO_4 + SO_3 \longrightarrow H_2S_2O_7$ (olum)
- $SO_3 + PCl_5 \longrightarrow POCl_3 + SO_2 + Cl_2$

SULPHURIC ACID (H_2SO_4)

Manufactured by contact process

Step I $S + O_2 \longrightarrow SO_2$

Step II $SO_2 + O_2 \longrightarrow SO_3$

Step III $SO_3 + H_2SO_4 \longrightarrow H_2S_2O_7$ (olum)

Step IV $H_2S_2O_7 + H_2O \longrightarrow 2H_2SO_4$

PROPERTIES

colourless, dense, oily liquid (96 - 98% pure)

$H_2SO_4 + H_2O \longrightarrow H_3O^+ + HSO_4^-$

$HSO_4^- + H_2O \longrightarrow H_3O^+ + SO_4^{2-}$

Conc. H_2SO_4 is strong dehydrating agent.

$H_2C_2O_4 \xrightarrow[-H_2O]{H_2SO_4} CO + CO_2$

$C_{12}H_{22}O_{11} \xrightarrow{H_2SO_4} 12C + 11H_2O$

Hot conc^n. H_2SO_4 is moderate oxidising agent.

$HO-SO_2-OH \xrightarrow{PCl_5} Cl-SO_2-OH + POCl_3 + HCl$

$\xrightarrow{2PCl_5} Cl-SO_2-Cl + 2POCl_3 + 2HCl$

SODIUM THIOSULPHATE $Na_2S_2O_3 \cdot 5H_2O$ (Hypo)

PREPARATION

i. $Na_2SO_3 + S \longrightarrow Na_2S_2O_3$

ii. $6NaOH + 4S \xrightarrow{\Delta} Na_2S_2O_3 + 2Na_2S + H_2O$

PROPERTIES

an antichlor used in photography etc.

$Na_2S_2O_3$

- Rxn with Ag (Salts ($AgNO_3$ / AgCl / AgBr / AgI)) $\longrightarrow Ag_2S_2O_3 \downarrow$ (white)
- $FeCl_3$ $\longrightarrow Fe(S_2O_3)_2$ (Pink)
- $AuCl_3$ $\longrightarrow Na_3[Au(S_2O_3)_2]$
- $CuCl_2$ $\longrightarrow Na_4[Cu_6(S_2O_3)_5]$

- Heating Effect: $Na_2S_2O_3 \longrightarrow 3Na_2SO_4 + Na_2S_5$
- As Antichlor: $Na_2S_2O_3 + 4Cl_2 + 5H_2O \longrightarrow 2NaHSO_4 + 8HCl$
- Reax^n with HCl: $S_2O_3^{2-} + H^+ \longrightarrow S\downarrow + SO_2 + H_2O$

GROUP - 17

HALOGEN

PROPERTIES

Exist in diatomic form.

F_2 (yellow gas) Cl_2 (yellow green gas) Br_2 (orange Liq.) I_2 (violet vapour solid)

$$2F_2(g) + 2H_2O(l) \longrightarrow 4H^+ + 4F^-(aq.) + O_2(g)$$

I_2 gives reverse rxn of flourine :-

$$4I^- + 4H^+ + O_2 \longrightarrow 2I_2 + 2H_2O(l)$$

$$X_2(g) + H_2O(l) \longrightarrow HX(aq) + HOX(aq) \quad (X = Cl, Br)$$

F_2 is the strongest oxidising agent.

FLOURINE (F_2)

PROPERTIES

1. Pale yellow → green gas, pungent odour, poisonous
2. $2F_2 + 2NaOH\ (dil) \longrightarrow OF_2(g) + 2NaF + H_2O$

   $2F_2 + 4NaOH\ (conc.) \longrightarrow O_2(g) + 4NaF + 2H_2O$
3. $2NH_3 + 3F_2 \longrightarrow N_2 + 6HF$
4. $SiO_2 + 2F_2 \longrightarrow SiF_4 + O_2$

   other Halogen + $NH_3 \longrightarrow NX_3$
5. $H_2S + F_2 \longrightarrow SF_6 + 2HF$
6. $O_2 + F_2 \xrightarrow[\text{discharge}]{\text{silent } e^-} O_2F_2$
7. Reacts with metals and non-metals

   $2Ag + F_2 \longrightarrow 2AgF$

   $C + 2F_2 \longrightarrow CF_4$

   $2Al + 3F_2 \longrightarrow 2AlF_3$

   $Si + 2F_2 \longrightarrow SiF_4$

Homocarbons are extremely inert.

PREPARATION

$CaF_2 + H_2SO_4 \longrightarrow CaSO_4 + 2HF$

$KF + HF \longrightarrow K[HF_2] \longrightarrow H_2 + F_2$

Kathode $2H^+ + 2e^- \longrightarrow H_2(g)$

Anode $2F^- \longrightarrow F_2 + 2e^-$

Anode of carbon must be free from graphite
there should be no moisture in the vessel.
F_2 cannot be prepared by e⁻lysis of NaF or KF.

CHLORINE Cl_2

PREPARATION

$4H^+ + MnO_2 + 2Cl \longrightarrow Cl_2 + Mn^{+2} + 2H_2O$

Deacon's Process :- $4HCL \quad O_2 \quad 2Cl_2 \quad 2H_2O$

Castner - Kellner Cell :- $NaCl \longrightarrow Na^+ + Cl^-$

Anode :- $2Cl^- \longrightarrow Cl_2 + 2e^-$



$+ 2HCL \longrightarrow CaCl_2 + Cl_2 + H_2O$

$NaCl + O_2 \longrightarrow 2NO_2 + Cl_2$

PROPERTIES

Green - yellow gas with suffocating smell.

Bleaching action - $Cl_2 + H_2O \longrightarrow HCL + HOCL$

$HOCL \longrightarrow HCL + [O]$

coloured subs + $[O] \longrightarrow$ colourless

oxidative & permanent bleaching using Cl_2/H_2O.

It act as an oxidising agent due to nascent.

Reaxn with NaOH -

$Cl_2 + NaOH$ (cold & dil) $\longrightarrow NaCl + NaOCl + H_2O$ [P]

Cl_2 NaOH (hot & conc) $NaClO_3$ NaCl H_2O

$2Ca(OH)_2 + 2Cl_2 \longrightarrow Ca(OCl)_2 + CaCl_2 + H_2O$

BROMINE (Br_2)

PREPARATION

$$NaBr + MnO_2 + H_2SO_4 \longrightarrow Bl_2 + Mn^{+2} + NaHSO_4 + H_2O$$

$$2Br^- + Cl_2 \longrightarrow 2Cl^- + Br_2$$

PROPERTIES

Reddish brown liquid.

IODINE (I_2)

PREPARATION

$$NaI + MnO_2 + H_2SO_4 \longrightarrow I_2 + MnSO_4 + NaHSO_4 + H_2O$$

$$I^- + IO_3^- + H^+ \longrightarrow I_2 + H_2O$$

$$CuSO_4 + 2KI \longrightarrow K_2SO_4 + CuI_2$$

PROPERTIES

Dark violet solid

$$S_2O_3^{2-} + I_2 \longrightarrow S_4O_8^{2-}$$ (Tetra thionate ion)

$$KClO_3 + I_2 \longrightarrow KIO_3 + Cl_2$$

HALOGEN ACIDS (HCl, HBr, HI)

PREPARATION

(i) By direct combination

$$H_2 + Cl_2 \longrightarrow 2HCl \qquad H_2 + Br_2 \longrightarrow 2HBr \qquad H_2 + I_2 \longrightarrow 2HI$$

(ii) By heating halide with conc. acid

$$NaCl + H_2SO_4 \longrightarrow NaHSO_4 + HCl$$

HCl cannot be dried over P_2O_5 or CaO, It is dried by passing through conc^n H_2SO_4.

$KX + H_2SO_4 \longrightarrow KHSO_4 + HX$ ($X = Br$ or I)

$H_2SO_4 + 2HX \longrightarrow SO_2 + X_2 + 2H_2O$

HBr and HI cannot be prepared by conc^n H_2SO_4 as they act as reducing agent

$3KX + H_3PO_3 \longrightarrow K_3PO_4 + 3HX$ ($X = Br, I$)

PROPERTIES

- Acidic strength HI HBr HCl HF
- Aqua regia contains HCl and HNO_3 (conc.) in 3:1 ratio

$Au + 4H^+ + NO_3^- + 4Cl^- \longrightarrow [AuCl_4] + NO + 2H_2O$

$3Pt + 16H^+ + 4NO_3^- + 18Cl^- \longrightarrow 3[PtCl_6] + 4NO + 8H_2O$

- HCl, HBr, HI have reducing property.

CHLORINE DIOXIDE (ClO_2)

PREPARATION

$2HClO_3 + 2HCl \longrightarrow 2ClO_2 + 2H_2O + Cl_2$

PROPERTIES

Yellow gas at room temperature

$ClO_2 \xrightarrow{H_2O} ClO + [O]$

$2ClO_2 + 2O_3 \longrightarrow Cl_2O_6$ (yellow solid) $+ 2O_2$

$Cl_2O_6(s) \longrightarrow [ClO_2] + [ClO_4]^-$

Cl_2O_6 is a mixed oxide of $HClO_3 + HClO_4$

BLEACHING POWDER

Calcium hypochlorite — mixed salt of HCl, HOCl

- Composition — $Ca(OCl)_2 \cdot CaCl_2 \cdot Ca(OH)_2 \cdot 2H_2O$
- Preparation — $Ca(OH)_2 + Cl_2 \longrightarrow Ca(OCl)Cl\ H_2O$
- Auto oxidation — $6CaOCl_2 \longrightarrow Ca(ClO_3)_2 + 5CaCl_2$
- Oxidising agent — $CaOCl_2 \longrightarrow CaCl_2 + [O]$
- Available chlorine — $CaOCl_2 + CO_2 \longrightarrow CaCO_3 + Cl_2$

SOME IMPORTANT ORDER

- Acidic strength
  - (i) $HI > HBr > HCl > HF$
  - (ii) $HOCl > HOBr > HOI$
- Oxidising power
  - (i) $F_2 > Cl_2 > Br_2 > I_2$
  - (ii) $BrO_4^- > IO_4^- > ClO_4^-$
- Order of disproportionation
  - (i) $3XO^- \longrightarrow 2X^- + XO_3^-$
  - (ii) $IO^- > BrO^- > ClO^-$

PSEUDO HALIDES

Inorganic compounds with two or more atoms of which atleast one is N, behave like Helogens.

E.g. cyanogen $(CN)_2$, $(SCN)_2$, $(SeCN)_2$, $(SCSN_3)_2$, (OCN^-), $(NCN)^{2-}$ $(N_3)^-$ etc.

INTERHALOGEN COMPOUNDS

- Due to difference in electronegativity, halogen atoms combine with each other to form interhalogen compound.
- TYPES - AB, AB_3, AB_5, AB_7 where A is less EN atom and B is more EN atom.
- PREPARATION

$$Cl_2 + F_2 \ (1:1) \longrightarrow 2ClF$$

$$Cl_2 + 3F_2 \longrightarrow 2ClF_3$$

$$I_2 + Cl_2 \longrightarrow 2ICl$$

- Interhalogen compounds containing flourine are colourless but those of other halogens are coloured.
- They all are covalent and diamagnetic in nature.
- Boiling pt. ↑ with ↑ in EN between A and B.
- Stability ↓ with ↓ in EN between A and B.

$IF > BrF > ClF > ICl > IBr > BrCl$ (stability)

- These are more reactive than parent halogens but less reactive than F_2.

$ClF_3 > BrF_3 > IF_7 > BrF_5 > BrF$ (reactivity)

- When hydrolysed, they form HB and HOA.

$$AB + H_2O \longrightarrow HB + HOA$$

E.g

$$BrCl + H_2O \longrightarrow HCl + HOBr$$

$$IF_7 + H_2O \longrightarrow HF + H_5IO_6$$

$$BrF_5 + 3H_2O \longrightarrow 5HF + HBrO_3$$

GROUP - ZERO

NOBLE GASES

PROPERTIES

They are least reactive in periodic table.
Have high I.P and more positive E.A.
Highest radii (vanderWall) in respective period.
first noble gas compound was $O_2^+ PtF_6^-$ (red).

Analog $\longrightarrow XePtF_6$ (red)

Inert gas molecules get teapped in the cages formed by crystal str. of H_2O leading to clathrate compound. He, Ne, Ar donot fourmed clatharate compounds due to small size.

XENON DIFLUORIDE (XeF_2)

PREPARATION

$Xe + F_2 \xrightarrow{(2:1)} XeF_2$

$Xe + O_2F_2 \longrightarrow XeF_2 \; O_2$

$2AgF_3 + Xe \longrightarrow 2AgF_2$ (Brown) $+ XeF_2$

PROPERTIES

(i) $XeF_2 + H_2 \longrightarrow Xe + 2HF$

(ii) $2XeF_2 + 2H_2O \longrightarrow 2Xe + 4HF + O_2$ (slow)

$XeF_2 + 2NaOH \longrightarrow Xe + 2NaF + H_2O + \frac{1}{2}O_2$ (fast)

(iii) Act as flouringating agent

$XeF_2 + PF_5 \longrightarrow [XeF]^+ \; [PF_6]^-$

$XeF_2 + IF_5 \longrightarrow [XeF]^+ \; [IF_6]^-$

XENON TETRAFLUORIDE (XeF_4)

PREPARATION

(i) $Xe + 2F_2 \longrightarrow XeF_4$ (1:5)

PROPERTIES

(ii) $6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$

(ii) $XeO_3 + NaOH \longrightarrow Na^+[HXeO_4]^-$ (sodium xenate)

(iii) It disproportionate in basic medium.

$2[HXeO_4]^- + 2OH^- \longrightarrow [XeO_6]^{4-} + Xe + O_2 + H_2O$

(xenate ion) (Perxenate ion)

$[XeO_6]^{4-} \xrightarrow[Water]{H^+} XeO_3 + O_2$

(iv) Act as fluorinating agent.

$XeF_4 + Pt \longrightarrow PtF_4 + Xe$

$XeF_4 + 4NO \longrightarrow Xe + 4NOF$ (nitrosyl fluoride)

$XeF_4 + 4NO_2 \longrightarrow Xe + 4NO_2F$ (nitronium fluoride)

$XeF_4 + 2C_6H_6 \longrightarrow C_6H_5F + 2HF + Xe$

XENON HEXAFLUORIDE (XeF_6)

PREPARATION

(i) $Xe \quad 3F_2 \quad (1:20) \quad XeF_6$

(ii) $XeF_4 \quad O_2F_2 \quad XeF_6 \quad O_2$

PROPERTIES

complete hydrolysis $XeF_6 + 3H_2O \longrightarrow XeO_3 + 6F$

Partial hydrolysis $XeF_6 + H_2O \longrightarrow XeOF_4 + 3HF$

$2XeF_6 \longrightarrow XeF_4 + 3F_2$

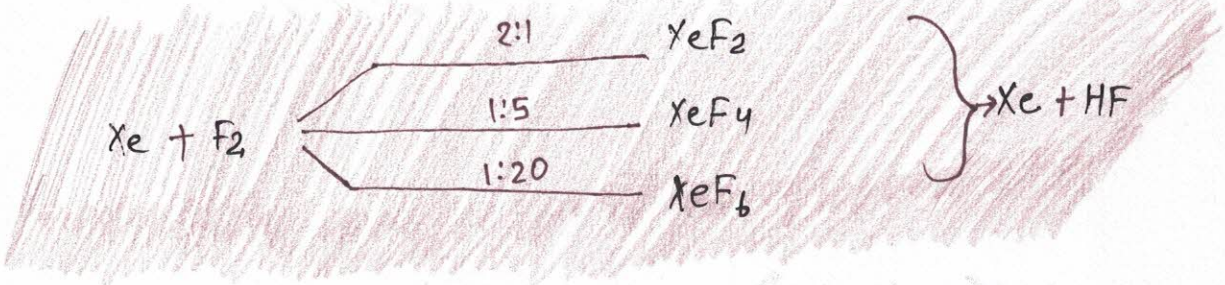
$XeO_3 + 2XeF_6 \longrightarrow 3XeOF_4$

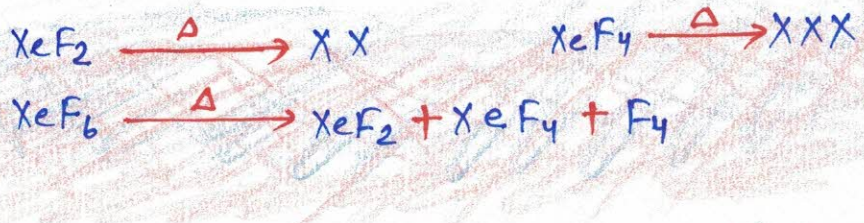
Fluorinating agent

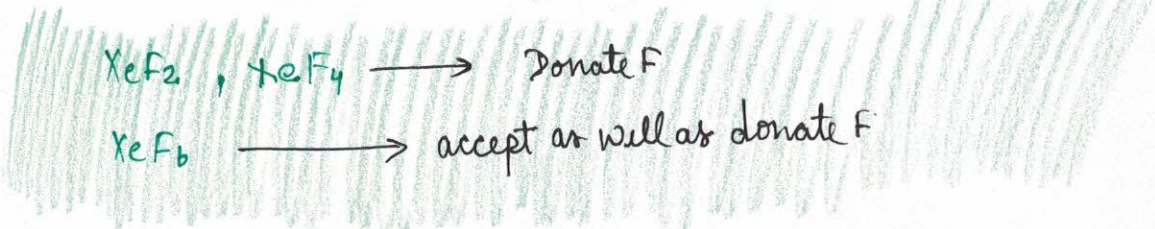
$CsF + XeF_6 \longrightarrow CsXeF_7$

$2CsXeF_7 + XeF_6 \longrightarrow Cs_2XeF_8$

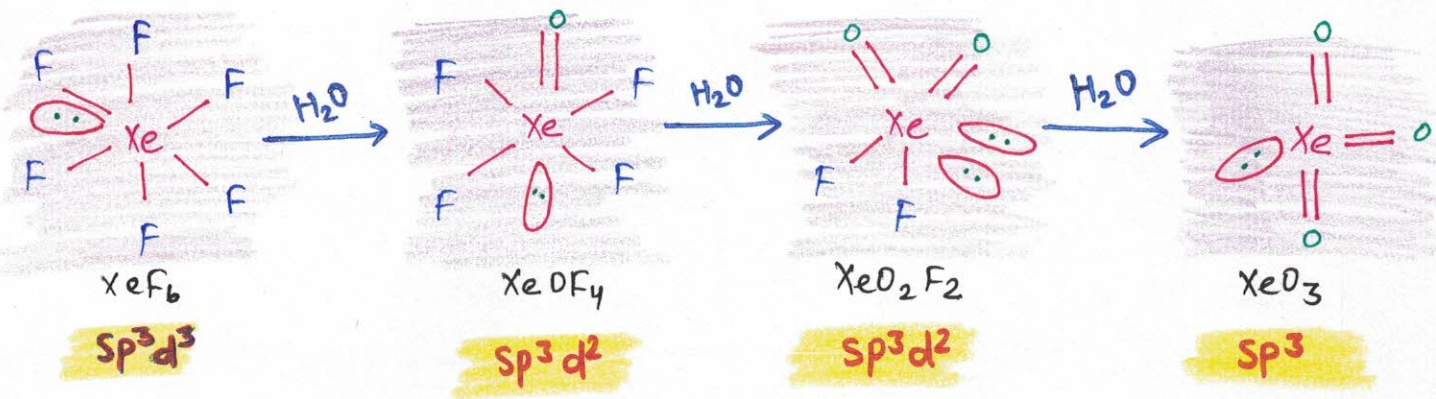
$XeF_6 + PtF_5 \longrightarrow (XeF_5)[PtF_6]^-$


Xe + F2
2:1
1:5
1:20
XeF2
XeF4
XeF6
Xe + HF


XeF2 Δ X X
XeF4 Δ X X X
XeF6 Δ XeF2 + XeF4 + F4


XeF2 , XeF4 → Donate F
XeF6 → accept as well as donate F

STRUCTURES


H2O
H2O
H2O
XeF6
XeOF4
XeO2F2
XeO3
Sp3d3
Sp3d2
Sp3d2
Sp3